

From Attention to Adaptation: A Short Review of Scaling Levers in Deep Learning

CameraReady Example

Worked example shipped with the CameraReady paper pipeline
<https://github.com/e96031413/SKILLS/tree/main/axiv-paper-writer>

Abstract—Deep learning did not scale because of one idea. It scaled because four separate constraints were removed in sequence: depth was made trainable, optimization was made robust to hand-tuned learning rates, supervision was decoupled from the downstream task, and adaptation was decoupled from the full parameter count. This short review traces those four levers through eight primary sources and states what each one traded away in return. It is a worked example for a gated paper-writing pipeline, not a comprehensive survey; the Limitations section says plainly what eight citations cannot support.

Index Terms—deep learning, scaling, transformers, pre-training, low-rank adaptation, survey

I. INTRODUCTION

Between 2015 and 2021 the working recipe for a deep learning system changed almost completely. A practitioner in 2015 designed an architecture for a task, trained it from scratch on that task’s labelled data, and tuned a learning rate schedule by hand. A practitioner in 2021 downloaded a pre-trained model, adapted a small fraction of its parameters, and rarely touched the optimizer at all.

That shift is often narrated as a single story about scale. It is more useful to read it as four independent constraints being removed, each by a specific piece of work, and each at a specific cost. Residual connections made depth trainable [1]. Attention made sequence modelling parallel [2]. Pre-training decoupled supervision from the downstream task. Low-rank adaptation decoupled the cost of specializing a model from its parameter count.

This review covers those four levers and nothing else. Reinforcement learning, multimodality, and the systems and hardware work that made any of it affordable are out of scope, and eight citations would not support claims about them.

II. ARCHITECTURE AND OPTIMIZATION

The first constraint was depth. Stacking more layers degraded training accuracy, not just test accuracy, which ruled out overfitting as the explanation. Residual connections reframed each block as learning a perturbation of the identity rather than a full transformation, and networks an order of magnitude deeper became trainable [1]. The cost was architectural commitment: the shortcut path constrains what a block may do to its input.

The second constraint was optimization. Per-parameter adaptive step sizes with bias-corrected moment estimates removed most of the hand-tuning that a schedule had previously required [3]. The cost was diagnostic: an optimizer that

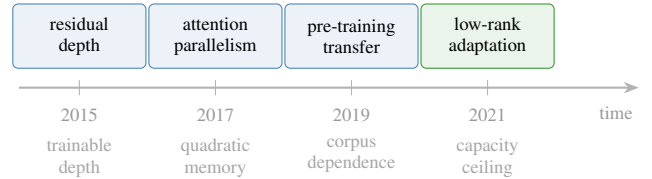


Fig. 1. The four levers, with what each one cost underneath. Dates are the first arXiv posting of the corresponding work. Drawn in TikZ rather than included as an image file, so the example compiles from a clean checkout with no external assets.

converges from a poor learning rate also converges quietly from a poor architecture, which makes a failure harder to localize.

The third constraint was sequential computation. Recurrence forced a sequence to be processed in order, so training time scaled with sequence length no matter how much hardware was available. Replacing recurrence entirely with attention made the computation parallel across positions [2]. The cost was quadratic: attention compares every position with every other, so memory grows with the square of the sequence length. Figure 1 places these three alongside the pre-training work discussed next.

III. PRE-TRAINING AND TRANSFER

Removing those three constraints made large models trainable. It did not make them useful, because a large model still needed a large labelled dataset for each task.

Masked language modelling broke that dependence. Training a bidirectional encoder to reconstruct masked tokens produces representations that transfer to tasks the model never saw, and the pre-training signal comes from unlabelled text [4]. The same recipe transfers across modalities: splitting an image into fixed-size patches and treating them as a token sequence lets an unmodified transformer encoder compete with convolutional architectures, provided the pre-training corpus is large enough [5]. Scaling the same objective further changes the interface rather than only the accuracy: at sufficient scale, a task can be specified in the prompt instead of through gradient updates [6].

The cost here is corpus dependence. Each of these results is a claim about a model *and* the corpus it was pre-trained on, and the two cannot be separated after the fact. Table I summarizes the four levers on that axis.

TABLE I
WHAT EACH LEVER CHANGED, AND WHAT IT COST.

Lever	Constraint removed	Cost incurred
Residual depth	Deeper networks stopped degrading during training	Each block is constrained to perturb its input
Attention	Sequence processing became parallel across positions	Memory grows quadratically with sequence length
Pre-training	Supervision decoupled from the downstream task	Results are inseparable from the pre-training corpus
Low-rank adaptation	Specialization decoupled from parameter count	Update capacity bounded by the chosen rank

IV. EFFICIENT ADAPTATION

Pre-training moved the cost rather than removing it. Adapting a pre-trained model by full fine-tuning produces a complete copy of its parameters per task, which becomes the binding constraint once a single model serves many tasks.

Constraining the weight update to a low-rank factorization removes that constraint: the frozen base model is shared, and each task stores only the small factors [7]. The cost is a capacity ceiling that the practitioner now chooses explicitly. The rank is a hyperparameter, and too low a rank silently caps how far the adapted model can move from the base.

The same decoupling appears outside language modelling. Diffusion models separate the generative process into many small denoising steps, each of which is easy to train, and pay for it in sampling cost [8]. The pattern across all four levers is the same shape: a hard constraint is exchanged for a tunable one.

V. DISCUSSION AND LIMITATIONS

Read as a sequence, the four levers are not four solutions. They are four relocations of the same difficulty. Depth became trainable and the constraint moved into the block structure. Sequences became parallel and the constraint moved into memory. Supervision became cheap and the constraint moved into the corpus. Adaptation became cheap and the constraint moved into a rank hyperparameter. Each step made the remaining constraint the one the practitioner chooses rather than the one the method imposes.

Limitations. This review synthesizes eight primary sources. That is enough to trace an argument and nowhere near enough to survey a field. It omits reinforcement learning from human feedback, multimodal architectures, mixture-of-experts routing, quantization, and the systems and hardware work without which none of these results would have been reachable. It reports no benchmark numbers, because we reproduced none of them, and it takes each paper’s own account of its contribution at face value rather than re-evaluating it. Readers should treat this as a map of one argument, not as coverage.

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